

On sub-Gaussian estimators of the mean

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FRAMEWORK

- * i.i.d sample in $\mathbb{R}$, $X_1, X_2, \dots, X_n$, the true mean, $\mu := EX_1$
- * the estimator, $\hat{\mu}_n = \hat{\mu}_n(X_1, \dots, X_n)$, $\hat{\mu}_n$ measurable
- * risk measure

MEAN SQUARE ERROR:

$$E|\hat{\mu}_n - \mu|^2$$

PROBABLY APPROXIMATELY CORRECT:

Set $\delta \in (0, 1)$ and find

$$\inf\{\varepsilon = \varepsilon(n, \delta) : P(|\hat{\mu}_n - \mu| > \varepsilon) \leq \delta\}.$$

Naive solution with $\hat{\mu}_n = \bar{\mu} = \sum_{i=1}^n X_i$:

$$\varepsilon = \sigma \sqrt{\frac{1}{n\delta}}$$

SUB-GAUSSIAN PERFORMANCE

Gaussian tails

$$P \left\{ |\bar{\mu}_n - \mu| > \frac{\sigma \Phi^{-1}(1 - \delta/2)}{\sqrt{n}} \right\} \rightarrow \delta$$

The trivial bound $1 - \Phi(x) \leq e^{-x^2/2}$ yields

$$\Phi^{-1}(1 - \delta/2) \leq \sqrt{2 \log(2/\delta)}.$$

CLT: Asymptotically

$$P \left\{ |\bar{\mu}_n - \mu| > \frac{\sigma \sqrt{2 \log(2/\delta)}}{\sqrt{n}} \right\} \leq \delta$$

SUB-GAUSSIAN PERFORMANCE

DEFINITION: $\hat{\mu}_n$ is *L-sub-Gaussian* if $\exists L > 0 : \forall n$ with probability at least $1 - \delta$

$$|\hat{\mu}_n - \mu| \leq \frac{L\sigma\sqrt{\log(2/\delta)}}{\sqrt{n}}.$$

THEOREM (DEVROYE, LERASLE, LUGOSI, OLIVEIRA; 2016).
Let $n > 5$, $\mu \in \mathbb{R}$, $\sigma > 0$ and $(2e^{-n/4}, 1/2)$. Then for any mean estimator $\hat{\mu}_n$, there exists a distribution with mean μ and variance σ^2 such that

$$P \left\{ |\hat{\mu}_n - \mu| > \sigma \sqrt{\frac{\log(1/\delta)}{n}} \right\} \geq \delta.$$

PERFORMANCE OF THE EMPIRICAL MEAN

A WELL KNOWN FACT:

$$\bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is optimal in case of Gaussian distribution. (Catoni, 2012)

SUB-GAUSSIAN DISTRIBUTION:

$$E \exp\{\lambda(X_i - \mu)\} \leq \exp\{\sigma^2 \lambda^2 / L^2\}$$

implies L-sub-Gaussian performance for the empirical mean, by the Chernoff bound - the criterion is equivalent to

$$(E|X_i - \mu|^p)^{1/p} \leq \sqrt{p}C.$$

PERFORMANCE OF THE EMPIRICAL MEAN

By Chebyshev's inequality implies the bound

$$P \left(|\bar{\mu}_n - \mu| \geq \sigma \sqrt{\frac{1}{n\delta}} \right) \leq \delta.$$

and this is essentially the best, (Catoni 2012):

For small enough δ and for arbitrary σ there exists a distribution for which

$$P \left(|\bar{\mu}_n - \mu| \geq \sigma \sqrt{\frac{1}{n\delta}} \left(1 - \frac{c\delta}{n} \right) \right) \geq \delta.$$

SOLUTION:

PERFORMANCE OF THE EMPIRICAL MEAN

Proof of the minimax bound: Consider the distribution with support $\{-n\eta, 0, n\eta\}$ defined by

$$P(\{n\eta\}) = P(\{-n\eta\}) = (1 - P(\{0\}))/2 = \frac{\nu}{2n^2\eta^2}.$$

This distribution has zero expectation and variance ν and

$$P(\bar{\mu}_n \geq \eta) = P(\bar{\mu}_n \leq -\eta) \geq P(\bar{\mu}_n = \eta) \geq \frac{\nu}{2n\eta^2} \left(1 - \frac{\nu}{n^2\eta^2}\right)^{n-1}.$$

MEDIAN-OF-MEANS ESTIMATOR

Nemirovsky, Yudin, "Problem complexity and method efficiency in optimization" (1983)

DEFINITION: The *median* of k real numbers $x_1, \dots, x_k$ is defined as $M(x_1, \dots, x_k) = x_i$ where x_i is such that

$$|\{j \in [k] : x_j \leq x_i\}| \geq \frac{k}{2} \quad \text{and} \quad |\{j \in [k] : x_j \geq x_i\}| \geq \frac{k}{2}.$$

Let $1 \leq k \leq n$ and partition $[n]$ into k blocks $B_1, \dots, B_k$, each of size $|B_j| \geq \lfloor n/k \rfloor \geq 2$. Given $X_1, \dots, X_n$, compute the sample mean in each block

$$Z_j = \frac{1}{|B_j|} \sum_{i \in B_j} X_i.$$

DEFINITION:

$$\hat{\mu}_n = M(Z_1, \dots, Z_k).$$

MEDIAN-OF-MEANS ESTIMATOR

By Chebyshev with probability equal or less then $1/4$,

$$|Z_j - \mu| > \sigma\sqrt{4/m}.$$

Note that $|\hat{\mu}_n - \mu| > \sigma\sqrt{4/m}$ implies that at least $k/2$ of the means Z_j are such that $|Z_j - \mu| > \sigma\sqrt{4/m}$. Hence, by Hoeffding's inequality

$$\begin{aligned} P \left\{ |\hat{\mu}_n - \mu| > \sigma\sqrt{4/m} \right\} &\leq P \{ \text{Bin}(k, 1/4) \geq k/2 \} \\ &= P \{ \text{Bin}(k, 1/4) - E\text{Bin}(k, 1/4) \leq k/4 \} \leq e^{-k/8} \end{aligned}$$

MEDIAN-OF-MEANS ESTIMATOR

Hsu, "Robust statistics" (2010)

THEOREM: Let $X_1, \dots, X_n$ be independent, identically distributed random variables with mean μ and variance σ^2 . Let m, k be positive integers and assume that $n = mk$. Then the median-of-means estimator $\hat{\mu}_n$ with k blocks satisfies

$$P \left\{ |\hat{\mu}_n - \mu| > \sigma \sqrt{4/m} \right\} \leq e^{-k/8}.$$

In particular, for any $\delta \in (0, 1)$, if $k = \lceil 8 \log(1/\delta) \rceil$, then, with probability at least $1 - \delta$,

$$|\hat{\mu}_n - \mu| \leq \sigma \sqrt{\frac{32 \log(1/\delta)}{n}}.$$

- * the median-of-means estimator has a sub-Gaussian performance with $L = 8$
- * the estimator depends on the confidence parameter

MEDIAN-OF-MEANS ESTIMATOR - RAMIFICATIONS

Theorem: Let $\alpha \in (0, 1]$ and let $X_1, \dots, X_n$ be independent, identically distributed random variables with mean μ and $(1 + \alpha)$ -th central moment $M = E[|X_i - \mu|^{1+\alpha}]$. Let m, k be positive integers and assume that $n = mk$. Then the median-of-means estimator with $k = \lceil 8 \log(2/\delta) \rceil$ blocks satisfies

$$P \left\{ |\hat{\mu}_n - \mu| > 8 \left(\frac{12 M^{1/\alpha} \log(1/\delta)}{n} \right)^{\alpha/(1+\alpha)} \right\} \leq \delta$$

Moreover, for any mean estimator $\hat{\mu}_n$, there exists a distribution with mean μ and $(1 + \alpha)$ -th central moment M such that

$$P \left\{ |\hat{\mu}_n - \mu| > \left(\frac{M^{1/\alpha} \log(2/\delta)}{n} \right)^{\alpha/(1+\alpha)} \right\} \geq \delta.$$

MEDIAN-OF-MEANS ESTIMATOR - RAMIFICATIONS

THEOREM: Let $X_1, \dots, X_n$ be independent, identically distributed random variables with mean μ and third central moment $\rho = E|X_i - \mu|^3$. Let m, k be positive integers and assume that $n = mk$. Assume that

$$\sqrt{\frac{\log(2/\delta)}{2k}} + \frac{\delta}{2\sigma^3\sqrt{m}} \leq 1/4.$$

Then the median-of-means estimator $\hat{\mu}_n$ with k blocks satisfy that, with probability at least $1 - \delta$

$$|\hat{\mu}_n - \mu| \leq \frac{1}{c} \left(\sigma \sqrt{\frac{\log(2/\delta)}{2n}} + \frac{\rho k}{2\sigma^2 n} \right),$$

where $c = \phi(\Phi^{-1}(3/4))$ is a constant.

MEDIAN-OF-MEANS ESTIMATOR - RAMIFICATIONS

OBSERVATIONS:

- * the second term is smaller then the first if

$$k \leq \frac{\sqrt{2}\sigma^3}{\rho} \sqrt{n \log(2/\delta)}$$

- * it suffices to set

$$k \leq \frac{\sqrt{2}\sigma^3}{\rho} \sqrt{n}$$

- * no dependence on the confidence parameter
- * sub-Gaussian performance for a wide range of δ
- * the block size can be as large as $C\sqrt{n}$ and thus δ can be in the range $(e^{-c\sqrt{n}}, 1)$
- * $k = o(\sqrt{n})$ then $\sqrt{n}(\hat{\mu}_n - \mu)$ is asymptotically normal

CATONI'S ESTIMATOR

NOTE:

$$\sum_{i=1}^n (X_i - y) = 0$$

Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing antisymmetric function and α a real parameter.

DEFINITION:

$$R_{n,\alpha}(y) = \sum_{i=1}^n \psi(\alpha(X_i - y))$$

Set

$$\psi(x) = \begin{cases} \log(1 + x + x^2/2), & x \geq 0 \\ -\log(1 - x + x^2/2), & x < 0 \end{cases}$$

Catoni's estimator $\hat{\mu}_{\alpha,n}$ is defined as the unique value y such that

$$R_{n,\alpha}(y) = 0.$$

CATONI'S ESTIMATOR

Using $\psi(x) \leq \log(1 + x + x^2/2)$ we have

$$\begin{aligned} E \left[e^{R_{n,\alpha}(y)} \right] &\leq \left(E \left[1 + \alpha(X - y) + \frac{\alpha^2(X - \mu)^2}{2} \right] \right)^n \\ &= \left(1 + \alpha(\mu - y) + \frac{\alpha^2(\sigma^2 + (\mu - y)^2)}{2} \right)^n \\ &\leq \exp \left(n\alpha(\mu - y) + \frac{n\alpha^2(\sigma^2 + (\mu - y)^2)}{2} \right) \end{aligned}$$

By Markov's inequality

$$P \left\{ R_{n,\alpha}(y) \geq n\alpha(\mu - y) + \frac{n\alpha^2(\sigma^2 + (\mu - y)^2)}{2} + \log(1/\delta) \right\} \leq \delta.$$

CATONI'S ESTIMATOR

Choose α so that $\alpha^2 \sigma^2 + 2 \log(1/\delta)/n \leq 1$. Then we have a root for

$$n\alpha(\mu - y) + \frac{n\alpha^2 (\sigma^2 + (\mu - y)^2)}{2} + \log(1/\delta),$$

consider the smaller one and denote it

$$y_+ = \mu + \frac{\frac{\alpha\sigma^2}{2} + \frac{\log(1/\delta)}{n\alpha}}{\frac{1}{2} + \frac{1}{2}\sqrt{1 - \alpha^2\sigma^2 - \frac{2\log(1/\delta)}{n}}}.$$

For this we have

$$P \{ R_{n,\alpha}(y_+) < 0 \} \geq 1 - \delta.$$

CATONI'S ESTIMATOR

Since $R_{n,\alpha}$ is strictly decreasing, this implies

$$P\{\hat{\mu}_{\alpha,n} < y_+\} \geq 1 - \delta,$$

and similarly an other constant

$$y_- = \mu - \frac{\frac{\alpha\sigma^2}{2} + \frac{\log(1/\delta)}{n\alpha}}{\frac{1}{2} + \frac{1}{2}\sqrt{1 - \alpha^2\sigma^2 - \frac{2\log(1/\delta)}{n}}}$$

can be constructed so that

$$P\{\hat{\mu}_{\alpha,n} > y_-\} \geq 1 - \delta.$$

CATONI'S ESTIMATOR

THEOREM: Let $X_1, \dots, X_n$ be independent, identically distributed random variables with mean μ and variance σ^2 . Let $\delta \in (0, 1)$ be such that $n > 2 \log(1/\delta)$. Catoni's mean estimator $\hat{\mu}_{n,\alpha}$ with parameter

$$\alpha = \sqrt{\frac{2 \log(1/\delta)}{n\sigma^2 \left(1 + \frac{2 \log(1/\delta)}{n - 2 \log(1/\delta)}\right)}}$$

satisfies that, with probability at least $1 - 2\delta$,

$$|\hat{\mu}_{n,\alpha} - \mu| < \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n - 2 \log(1/\delta)}}.$$

TRIMMED-MEAN ESTIMATOR

A VARIANT:

- * sample: $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$
- * $\phi_{\alpha, \beta}(x) = \beta$, $x > \beta$ and $x \in [\alpha, \beta]$ and α , $x < \alpha$
- * denote the rearrangement as $Y_1^*, \dots, Y_n^*$

DEFINITION: Given the confidence level $\delta \geq 8e^{-3n/16}$, set

$$\varepsilon = \frac{16 \log(8/\delta)}{3n}.$$

Let $\alpha = Y_{\varepsilon n}^*$ and $\beta = Y_{(1-\varepsilon)n}^*$ and set

$$\hat{\mu}_{2n} = \frac{1}{n} \sum_{i=1}^n \phi_{\alpha, \beta}(X_i).$$

TRIMMED-MEAN ESTIMATOR

THEOREM: Let $X_1, \dots, X_n, Y_1, \dots, Y_n$ be independent, identically distributed random variables with mean μ and variance σ^2 . Let $\delta \in (0, 1)$ be such that $n > (16/3) \log(8/\delta)$. Then, with probability at least $1 - \delta$,

$$|\hat{\mu}_{2n} - \mu| \leq 9\sigma \sqrt{\frac{\log(8/\delta)}{n}}.$$

* robust to adversarial contamination:

Lugosi, Mendelson: Robust multivariate mean estimation: the optimality of trimmed mean.

$$P \left\{ |\hat{\mu}_{2n} - \mu| \leq c\sigma \left(\sqrt{\eta} + \sqrt{\frac{\log(1/\delta)}{n}} \right) \right\} \geq 1 - \delta$$

NOTE ON MULTIPLE- δ ESTIMATORS

THEOREM: For every $L \geq 50$ for every sample size n , no estimator can be simultaneously L -sub-Gaussian for both $\delta_1 = 1/(2e\sqrt{L^3 + 1})$ and $\delta_2 = 2e^{-L^4/4}$ for all distribution with finite second moment.

PROOF.

$$\begin{aligned} & P \left\{ \hat{\mu}_n(Y_1, \dots, Y_n) > z \mid \sum_{i=1}^n Y_i = c \right\} \\ &= P \left\{ \hat{\mu}_n(X_1, \dots, X_n) > z \mid \sum_{i=1}^n X_i = c \right\} \end{aligned}$$

TYPICAL ADDITIONAL ASSUMPTIONS:

- * upper and lower bounds on the variance
- * existence of higher moments
- * symmetry

MULTIVARIATE SCHEME

SETUP:

- * X takes values in $\mathbb{R}^d$
- * $\mu = E[X]$, $\Sigma = E(X - \mu)(X - \mu)^T$
- * sample drawn from $X : X_1, \dots, X_n$
- * task: estimate the vector μ

$$P(\|\hat{\mu}_n - \mu\| > r) \leq \delta$$

The naive approach yields:

$$r = \sqrt{\frac{\text{Tr}(\Sigma)}{n\delta}}.$$

SUB-GAUSSIAN PERFORMANCE - THE NORMAL CASE

Let the sample be multivariate normal. Then the sample mean $\bar{\mu}_n \sim \mathcal{N}_d(\mu, \frac{1}{n}\Sigma)$ Setting $\bar{X} \sim \mathcal{N}_d(0, \Sigma)$ we have

$$\begin{aligned} & P\left(\|\bar{\mu}_n - \mu\| \geq \sqrt{\frac{\text{Tr}\Sigma}{n}} + t\right) \\ & \leq P(\|\bar{\mu}_n - \mu\| \geq E\|\bar{\mu}_n - \mu\| + t) \\ & = P(\|\bar{X}\| - E\|\bar{X}\| \geq t\sqrt{n}) \end{aligned}$$

SUB-GAUSSIAN PERFORMANCE - THE NORMAL CASE

A key property of $\bar{X}$ is that $\bar{X} \sim \Sigma^{1/2}Y$, where $Y \sim \mathcal{N}_d(0, I_d)$ and we also have

$$\left| \|\Sigma^{1/2}x\| - \|\Sigma^{1/2}y\| \right| \leq \|\Sigma^{1/2}(x - y)\| \leq \|\Sigma^{1/2}\|_{2 \rightarrow 2} \cdot \|x - y\|.$$

and a concentration inequality of Tsirelson, Ibragimov, Sudakov yields

$$P\left(\|\bar{X}\| - E\|\bar{X}\| \geq t\sqrt{n}\right) \leq e^{-nt^2/(2\lambda_{\max})}.$$

Tsirelson, Ibragimov, Sudakov, "Norm of gaussian sample function"

SUB-GAUSSIAN PERFORMANCE - DEFINITION

DEFINITION:

We say that $\hat{\mu}_{n,\delta}$ has *sub-Gaussian* performance if for $\delta \in (0, 1)$ with probability at least $1 - \delta$ we have

$$\|\hat{\mu}_{n,\delta} - \mu\| \leq \sqrt{\frac{\text{Tr}(\Sigma)}{n}} + \sqrt{\frac{2\lambda_{\max} \log(1/\delta)}{n}}$$

Using the findings of the previous slide yields that $\bar{\mu}_n$, in case of a normal sample, is sub-Gaussian.

SUB-GAUSSIAN PERFORMANCE - HISTORICAL RESULTS

- * If the data has *sub-Gaussian distribution*, that is

$$E \exp\{\lambda \langle v, X - EX \rangle\} \leq \exp\{c\lambda^2 \langle v, \Sigma v \rangle\}$$

then the sample mean is sub-Gaussian in the above sense.

- * the geometric median estimator $\tilde{\mu}_{n,k}$, the point that minimizes the sum of the Euclidean distances to the sample means within each block, gives with probability at least $1 - \delta$

$$\|\tilde{\mu}_{n,k} - \mu\| \leq C \sqrt{\frac{\text{Tr}(\Sigma) \log(1/\delta)}{n}}.$$

- * an other estimator, denote it $\hat{\mu}_n^{(\delta)}$, produced by (Joly, Lugosi, Oliveira), with an additional fourth moment assumption

$$E [\langle (X - \mu), v \rangle^4] \leq K(\langle v, \Sigma v \rangle)^2$$

and with $n \geq CK \log(d)(d + \log(1/\delta))$ gives with probability at least $1 - \delta$

$$\|\hat{\mu}_n^{(\delta)} - \mu\| \leq C \left(\sqrt{\frac{\text{Tr}(\Sigma)}{n}} + \sqrt{\frac{\lambda_{\max} \log((1/\delta) \log(d))}{n}} \right)$$

MULTIVARIATE MEDIAN-OF-MEANS

GEOMETRIC MEDIAN:

$$\tilde{\mu}_{n,k} = \operatorname{argmin}_{m \in \mathbb{R}^d} \sum_{j=1}^k \|Z_j - m\|$$

MULTIVARIATE MEDIAN:

$$S_a = \{x \in \mathbb{R}^d :$$

$$\exists J \subset [k], |J| \geq k/2 \text{ such that } j \in J, \min_{j \in J} (\|Z_j - x\| - \|Z_j - a\|) > 0\}$$

$$\hat{\mu}_n \in \operatorname{argmin}_{a \in \mathbb{R}^d} \operatorname{diam}(S_a^c)$$